

GENERAL SPECIFICATION G1.2

Information Management

CONTENTS

G1.2	Information Management.....	1
G1.2.1	Overview	1
G1.2.2	Common Data Environment	1
G1.2.3	Electronic Document Management System	2
G1.2.4	Approval of Submitted Content	3
G1.2.5	Contractor's CDE	4
G1.2.6	Document Control and BS-1192 Work Processes.....	4
G1.2.7	File Naming Convention.....	4
G1.2.8	Occasional Hard Copy Submissions	5
G1.2.9	Non-Graphical Data.....	5
G1.2.10	P&ID Data.....	5
G1.2.11	BIM Execution (Graphical Content)	6
G1.2.12	BIM Execution Plan	9
G1.2.13	Model Submissions during Construction	10
G1.2.14	Model Submissions After Construction (As-Built)	10
G1.2.15	Integration with Asset Management System (AMS).....	11
G1.2.16	Facility Management (FM) and Operations & Maintenance (O&M)	12
G1.2.17	Information Management Roles.....	13

G1.2 INFORMATION MANAGEMENT

G1.2.1 Overview

- (a) The TWRP project will be delivered using Building Information Management (BIM) principles as defined by BS 1192 and the Singapore BIM standard when providing the Works. This section describes the Board's Information Requirements (BIR). The Contractor shall comply with the BIR described herein. The S.O. will maintain the BIR and shall communicate any changes or updates to the Contractor as the Works progress.
- (b) The BIR also applies to the Contractor's supply chain. The Contractor shall ensure that all procurement contracts include the BIR for the Project and that the supply chain provides information in the correct format.
- (c) Exceptions to any aspect of the BIR, shall be subject to the agreement of the S.O. and where agreed, are to be highlighted in the Contractor's BIM Execution Plan (BEP).
- (d) The BIM principles to be followed for the TWRP project are based on BS 1192: 2007 and PAS 1192. An Information Model comprises Documentation, Graphical and Non-Graphical data which is enriched over time as indicated in **Figure 1**.

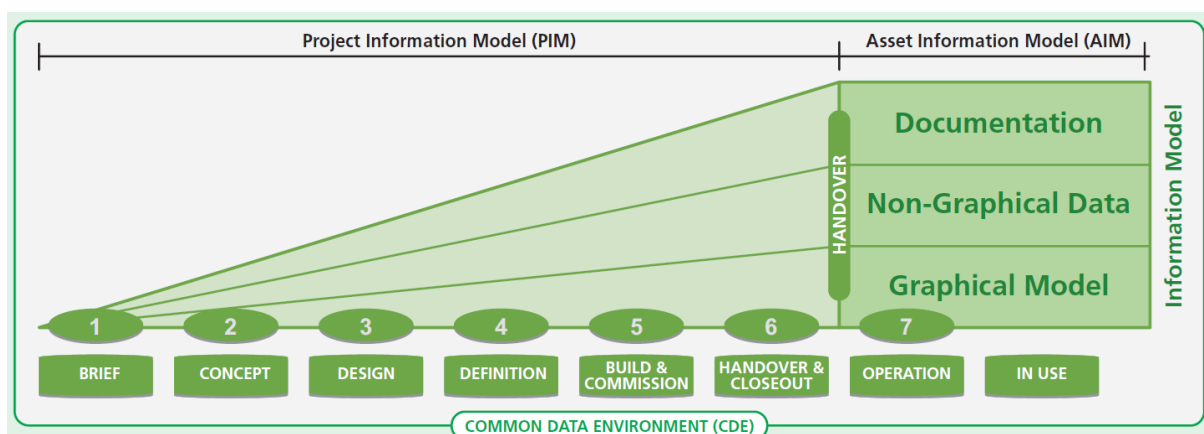


Figure 1: Definition of Information Model in a Common Data Environment

- (e) The Information Model shall be housed within a Common Data Environment (CDE), which can be a collection of systems, which together provide the functionality required to meet BS 1192.
- (f) The BIR defines the requirements for the production and management of documentation and non-graphical content for the project as well as the Common Data Environment.

G1.2.2 Common Data Environment

- (a) The Contractor shall maintain a Common Data Environment (CDE), to store and manage all information relevant to the Contract.

- (b) The Contractor's CDE shall:
 - (i) exchange information with the S.O.'s project wide CDE, referred to as the TWRP CDE. Information transfer shall be via the S.O.'s Project Management Information System (TWRP PMIS); and
 - (ii) include an electronic document management system (EDMS) for management of all documentation transfers between the Contractor and the S.O..

G1.2.3 Electronic Document Management System

- (a) The Contractor shall comply with the requirements and work processes of the project-wide EDMS which will be provided by the S.O. to the Contractor upon Contract award. The project-wide EDMS will serve as a single digital platform for document exchange including but not limited to:
 - (i) Formal correspondences and submissions;
 - (ii) Monthly reports;
 - (iii) Deliverables;
 - (iv) Requests for information (RFI);
 - (v) Programme for the Works;
 - (vi) BIM Models;
 - (vii) SQL data;
 - (viii) Drawings;
 - (ix) Manufacturer material data;
 - (x) Authority submissions; and
 - (xi) Other relevant information.
- (b) The project-wide EDMS will be used by the S.O. for the management of contractual correspondence (including review and approval of Contractor submissions) and for the storage of all such permanent records of the project. The Contractor shall ensure that e-mail is not used for formal contract correspondence between the S.O. and the Contractor. Email may still be used for informal correspondence. However, informal correspondence shall a) not deal with submissions required under the Contract and b) not deal with any contractual matter.
- (c) The project-wide EDMS is a web-based application allowing electronic creation, transmittal and tracking of communications and notifications between the parties. Therefore, the Contractor is expected to ensure reliable access to the internet using hardware with sufficient internet bandwidth to allow efficient document exchange.
- (d) The S.O. will make available limited number of project-wide EDMS user licences for the Contractor which shall be used by Contractor to manage the document exchange with the S.O. and submissions related to TWRP project only. The Contractor shall notify the S.O. of the names of Contractor's nominated staff to be trained in the use of the project-wide EDMS and shall make them available for training as agreed with the S.O. The Contractor's nominated staff shall be full time for the duration of the Contract and are to attend meetings hosted by the S.O.

Information regarding user accounts, access and procedures for usage of the project-wide EDMS will be provided by the S.O..

- (e) All files within the project-wide EDMS shall remain the ownership of the Board. Notwithstanding the above, the Contractor shall employ his own PMIS to interface with the project-wide EDMS and to manage information flow between the Contractor and supply chain.

G1.2.4 Approval of Submitted Content

- (a) The Contractor shall submit all items to the S.O. using the project-wide EDMS. Approval by the S.O. of a submission by the Contractor shall mean that the S.O. has no objection in principle to the content of the submission, and that the S.O. finds the content of the submission good enough to allow the work relating to the submission to proceed. Approval by the S.O. shall not relieve the Contractor of his responsibility under the Contract. Approval will be given based on information made available at the time to the S.O. on the understanding that subsequent events may require withdrawal or modification of such approval.
- (b) On completion of review, the S.O. will respond to the submission with one of the following comments:
 - (i) "Approved" : to allow work to proceed;
 - (ii) "Approved subject to..." : to allow work to proceed subject to stated amendments and conditions; and
 - (iii) "Not Approved" : revised submission required to include stated requirements.
- (c) Submittals with "Not Approved" and "Approved subject to" statuses shall be revised and re-submitted within a maximum of 14 working days from the date of return.
- (d) Subject to any specific requirements in the Contract, all submissions shall be made sufficiently early to allow the S.O. time for review and approval, and for the Contractor to provide and submit revisions or alternatives.
- (e) Unless otherwise specified or agreed before submission, the Contractor shall allow not less than 20 working days excluding weekends and public holidays for review by the S.O. of designs, documents and samples which are required to be approved and 14 working days for review of re-submissions before manufacture or construction commences as part of the Permanent Works. The Contractor shall have demonstrated in his submissions that he has coordinated submissions by his subcontractor or suppliers to confirm that the Contractor is satisfied that they are acceptable for incorporation into the Works.
- (f) If the S.O. does not respond to any submission within the specified number of days, then the Contractor may proceed with the work but shall inform the S.O. in writing that he is proceeding with the work. Absence of response from the S.O. will not relieve the Contractor of his obligations to meet all the requirements of the Specification.
- (g) All submissions shall be made sufficiently early to allow the S.O. time for review and approval of drawings, documents and samples which are required to be approved before commencement of manufacture of plant or of installation or construction on site as required, and for the Contractor to provide and submit revisions or

alternatives, or carry out further tests if, in the opinion of the S.O., the initial submissions do not meet the requirements of the Contract.

- (h) It shall be the responsibility of the Contractor to ensure that all relevant information is provided to enable the S.O. to easily determine whether a submitted design complies with the specific requirements or intent of the Specification. Where the S.O. assigns an inappropriate status due to lack of clarity or incomplete or incorrect information submitted by the Contractor, this shall not relieve the Contractor of his obligations to provide the Works in compliance with the Contract. Any costs for abortive works or redundant equipment provided because of such situations shall be borne by the Contractor.
- (i) The Contractor shall carry out the Works only in accordance with designs, models, drawings and documents, which have received the S.O.'s written approval.

G1.2.5 Contractor's CDE

- (a) The Contractor's CDE shall:
 - (i) Serve as a single source of information used to share contract information with the S.O.;
 - (ii) Be protected against cyber-attacks through compliance with Multi-Tier Cloud Security Singapore standard level 3 for cloud based solutions. The Contractor shall define the tier to be applied and provide the necessary evidence of applicable tier in the BIM Execution Plan prior to implementation alongside the proposed solution;
 - (iii) Include a Contractor's PMIS, which shall allow data to be backed up at scheduled intervals to a secure server for recovery; and
 - (iv) Enforce BS-1192 Work Processes for the Contractor and the supply chain.

G1.2.6 Document Control and BS-1192 Work Processes

- (a) The Contractor shall ensure that document control performed for the Contract is in alignment with the document control processes (including suitability and revision control), and File Naming Convention required by the S.O.. This includes the BS-1192 Work Process defined for document transmission and receipt, the management, control and storage of official documents, deliverables and correspondence produced for the Project, such as emails, letters, invoices, technical and progress reports, schedules, cost estimation, minutes of meetings, calculations, data sheets, bills of quantities, BIM Models, SQL data, drawings, etc.
- (b) The Contractor shall appoint dedicated person(s) to be responsible for the role of document control, and establish a Document Transmission Distribution Matrix (including approvers, reviewers etc). This matrix is to be used for when receiving; and before distributing, documents through the project-wide EDMS. This matrix is to be made available to the S.O. within four (4) weeks of the award of the Contract.

G1.2.7 File Naming Convention

- (a) All Contractors/Sub-Contractors shall adhere to a File Naming Convention provided by the S.O. upon Contract award which will be based on BS 1192.

G1.2.8 Occasional Hard Copy Submissions

- (a) From time to time, hard copies of documents may be necessary, for transmittal notes accompanying material sample submissions for example. Under such circumstances, transmittals shall be made via the project-wide EDMS with duplicate hard copies accompanying samples. The extent of hard copies is to be minimized. This shall be agreed between the Contractor and the S.O..

G1.2.9 Non-Graphical Data

- (a) The Contractor shall utilise the defined non-graphical data content which shall include, but shall not be limited to the following:
 - (i) BS 1192 Status, Revision, Workflow;
 - (ii) Document numbering;
 - (iii) Drawing numbering;
 - (iv) 4D sequence codes;
 - (v) 5D quantity take-off (QTO) codes;
 - (vi) 6D attributes;
 - (vii) Process-mechanical tagging codes;
 - (viii) Non- process-mechanical tagging codes;
 - (ix) Electrical tagging codes;
 - (x) Classification codes;
 - (xi) Interface lists; and
 - (xii) Material/Service/Fluid Codes.
- (b) The codes will be made available to the Contractor upon award. Management of the codes shall remain with the S.O.. Any changes or amendments requested to any of the code lists shall be administered by the S.O..

G1.2.10 P&ID Data

- (a) P&ID data is held in multiple Structured Query Language (SQL) databases and will be made available to the Contractor early in the Contract Period. The databases will be synchronised with P&ID drawings and shall be kept up to date by the Contractor with any Contractor changes. The P&ID database shall also be kept up to date with any Contractor-designed elements. The Contractor shall complete the database prior to handover of the asset.
- (b) The Contractor is expected to have suitably qualified persons to administer an SQL database and provide interrogation, issue resolution and exporting of content from SQL. The Contractor shall submit the SQL database along with the As-Built Model.

G1.2.11 BIM Execution (Graphical Content)

- (a) The Information Model will be the sole source from which 2D drawings, 3D models, 4D construction sequences, 5D quantities and cost tracking and 6D asset management deliverables are to be derived. The Contractor shall create models that are both complete and accurate, in accordance with this specification and with the Contractor's approved BIM Execution Plan.
- (b) In preparing the tender documentation, the S.O. has developed BIM models for civil, structural and MEICA work.
- (c) The Contractor shall be fully responsible for the following:
 - (i) For elements of the Works for which the Contractor is to complete the design (e.g. MEICA), the Contractor shall update the model with approved designs and provide an as-built model along with the linked database upon completion;
 - (ii) For elements of the Works for which the S.O. completes design (e.g. structural), the Contractor is to maintain the BIM model and update the model with as-built data;
 - (iii) For any scope or design changes, the Contractor shall update the model; and
 - (iv) Any temporary works which include but are not limited to vehicle and crane movements, temporary facilities, temporary roads, temporary works, formworks, scaffoldings those activities which impact any other Contract or require permits. A model may also be required to demonstrate that an activity is safe and/or does not impact other Contractors or within contracted work.
 - (v) Model temporary structure as per supplier shop drawings, details and method statements. Follow the BCA's BIM Essential guide for collaborative virtual design and construction
- (d) All models shall be compliant with the specifications. Taking the approach also that "if it is to be constructed it needs to be modelled."
- (e) The main objectives of using BIM to deliver the Works and the high-level methodology for achieving each objective are given in Table G1.2-1.

Table G1.2-1: BIM Objectives and Methodology

Objective	Methodology
Enhance ability to make informed and timely decisions and communicate with all stakeholders.	Develop model visualizations and intelligent data to support decision making process and present information in a fashion that is easily understood
Establish a common working environment for the project team to collaborate and succeed.	Enhance collaboration and communication to improve the quality of models and ensure timely project implementation.
Mitigate project risk to the Board and the Contractor by eliminating	Drive the process of information and model management to eliminate cost overruns at the project site where coordination is exponentially more expensive than the virtual environment.

Objective	Methodology
disconnected data and inaccurate models.	
Increase confidence and predictability of project expenditure and programme for the Works.	Integrate intelligent model data with project expenditure and programming activities.
Provide continual oversight of actual construction progress vs. virtually planned work.	Simulate complex construction sequences to identify and communicate any potential risks virtually, first.
Control and streamline information for seamless turnover and best in class Asset Management environment	Create a controlled data environment throughout the life cycle of the project to facilitate a digital exchange of operations and maintenance information at project completion. Mitigate the painful process of manual information input at the close of the programme. Eliminate the risk of maintenance and operations items that may require early attention while information is being input.
Enhance worker safety and productivity due to better coordinated information	Align to the mission of the Board, BCA and MoM to enhance safety and increase productivity to eliminate waste and deliver the Works efficiently.
Use new and innovative technologies where appropriate	Analyse the usefulness of innovative technologies that can be applied throughout the project to deliver Project objectives.

- (f) BIM Authoring Tools - Design/modelling Software to be used by the Contractor to deliver the Works shall be as described in Table G1.2-2.

Table G1.2-2: Mandatory BIM Authoring Tools

Design/Modelling Software	Version	Function
Primavera	P6	Construction Scheduling
Microsoft Office	2016	All documentation
SQL Server	TBC	Hosting P&ID database
Bentley Aecosim V8i	SS6	Architectural, Structural +6D COBie
Bentley Aecosim V8i	SS6	MVAC, Electrical, Plumbing, Fire Protection +6D COBie
Bentley Open Plant Modeller V8i	SS6	Process Mechanical, Instrumentation & Controls

Design/Modelling Software	Version	Function
Bentley Open Plant P&ID	SS6	P&ID - Process Mechanical, Instrumentation & Controls
Bentley InRoads	SS4	Civil - Site Grading, Roads, Tunnel, Yard Piping, Utilities, Geotech and Site Analysis
Bentley Navigator	V8i and CONNECT	2D Online Design Collaboration/Redlining, 3D Design Collaboration/Clash Detection + 4D Sequencing, +5D QTO + 6D Cobie
Adobe Acrobat	11.0+	2D Design Collaboration/Redlining
Autodesk NavisWorks	2017	Delivery of model to the Board/PM + 4D Sequencing + 5D QTO

- (g) Throughout the project it is understood that software version will change. The S.O. will evaluate newer version and determine if software version can be upgraded to take advantage of added features, and or functionality that might increase efficiencies for the project.
- (h) Alternative software for each of the above functions shall NOT be used. Software for additional functions such as design verification, reinforcement design etc. shall be proposed by the Contractor for the approval of the S.O..
- (i) At the time of Tender, the S.O. has developed BIM models of the Project using the design/modelling software given in Table G1.2-2. A set of modelling guidelines for BIM elements at different stages of the project are provided in the latest version of the Singapore BIM Guide (refer to <https://www.bca.gov.sg/bim/bimlinks.html>) and shall be referenced. In general, each element shall be modelled according to its size, shape, location, orientation and quantity. At the tender stage of the project, element properties are necessarily approximate. The Contractor shall develop element accuracy as the project progresses, until the level of accuracy is sufficient to support construction and installation.
- (j) The Contractor shall comply with the requirements of BCA for statutory submissions to regulatory agencies, as mandated by the prevailing BCA BIM e-submission regulations. The Contractor shall be responsible for liaison with BCA and other relevant regulatory agencies on all consultations and submissions related to the BIM development.
- (k) Architectural, Structural and Mechanical, Electrical, Plumbing (MEP) modelling guidelines and templates for Singapore BIM e-submissions for regulatory purposes can be found at: [https://www.corenet.gov.sg/general/building-information-modeling-\(bim\)-e-submission.aspx](https://www.corenet.gov.sg/general/building-information-modeling-(bim)-e-submission.aspx).

G1.2.12 BIM Execution Plan

- (a) The Contractor shall provide a fully-detailed BIM Execution Plan (BEP) for execution of the Works. The BEP shall describe the process in which BIM will be implemented throughout the Project lifecycle in compliance with this specification.
- (b) The BEP shall be updated and implemented in accordance with the requirements detailed herein. As the project progresses, any modifications to the objectives, project members, Model Authors, Model Users or basic BIM/CAD deliverables shall be agreed with the S.O. and documented in the BEP.
- (c) The BEP shall include the following information as a minimum:
 - (i) Project information - general project information, the project team, key personnel contact details and contract scope/ summary information;
 - (ii) Task information delivery plans (TIDP) - for each submission identified in the Programme for the Works, including a detailed information delivery schedule and register containing all identified information deliverables, coordination items and any supporting data or information; The TIDP is defined in PAS 1192;
 - (iii) BIM objectives and goals - BIM/CAD deliverables shall include models that describe the physical, functional and performance characteristics of the Works for visualization, simulation, analysis, collaboration, planning and documentation throughout the project lifecycle;
 - (iv) BIM quality control plan;
 - (v) Model development approach – the model shall be progressively developed by the Contractor as the Works progresses. The S.O.'s tender model shall be developed to become a virtual construction model, and then as an as-built model which will be handed over to the Board at the time of Substantial Completion;
 - (vi) BIM Uses – The Contractor shall define the BIM uses for the model and methodology to achieve those uses. These shall include but not be limited to Construction Sequencing (4D), Cost estimation (5D), Asset Management data turnover (6D);
 - (vii) BIM protocols - The Contractor shall identify and record, in each BEP, the BIM protocols used to create, manage and deliver the BIR;
 - (viii) Governance, roles and responsibilities - the Contractor shall identify in each BEP, information management roles as outlined in PAS 1192;
 - (ix) Procedures, methods and training - the Contractor shall identify in each BEP, the methods, procedures and processes related to the BIM uses and protocols;
 - (x) BIM deliverables – The Contractor shall confirm the models that will be delivered, and the Level of Detail and Level of Information;
 - (xi) Interface points – How the contractor will use the model to help interfacing with other contractors; and
 - (xii) Supply chain – Define how the contractor will engage their suppliers to ensure that they are fully engaging with the BIR.
- (d) The content of each of these sections, and all other necessary BEP sections, shall be developed by the Contractor and shall be subject to the approval of the S.O..

G1.2.13 Model Submissions during Construction

The Contractor shall provide information updates every month during execution of the Works. This shall include as a minimum (as appropriate as the Works progress).

- (a) Models including:
 - (i) 3D - the updated model including all approved designs, all construction works completed or in progress, all approved changes;
 - (ii) 4D - the model(s) in i.DGN and .imodel format with embedded Programme for the Works. This shall be a six month look ahead updated on a rolling monthly cycle; the model shall include temporary works, planned permitting works, interfaces with any other contractors as a minimum;
 - (iii) 5D - the model(s) in i.DGN and .imodel format with associated exports of data for project expenditure tracking and for verification of monthly progress claims;
 - (iv) 6D – updated models with as-built information embedded for each section of work completed that month;
 - (v) Clash resolution including clash detection reports;
 - (vi) A completed preparation and validation checklist for each model that the Contractor shall develop and provide;
 - (vii) Supporting CAD (2D) information in i.DGN format or as agreed in the BEP; and
 - (viii) Supporting or other required documentation in PDF format or as agreed in the BEP.
- (b) 2D denotes drawings in Adobe PDF. The Contractor shall provide PDF soft copies of all drawing sheets for review/comment by the S.O..
- (c) 3D denotes 3D models created using the BIM Applications required for the Works (Refer Table 2-2). The Contractor shall provide the BIM Models in both native file formats (.dgn), imodels (.i.dgn and .imodel) as well as Autodesk Navisworks (.NWD) format.
- (d) 4D denotes construction sequence/programme information. The Contractor shall embed Primavera (P6) scheduling data into the Navisworks Models. This process will require mapping 3D objects to the date/time as shown in the P6 Programme for the Works.
- (e) 5D denotes quantification and project expenditure tracking. The Contractor shall provide Navisworks model with quantification schedules attached.
- (f) 6D denotes asset management. Prior to completion of construction, the Contractor shall embed Construction Operations Building Information Exchange (COBie) data on all model elements that represent “Managed Assets”. Such data shall be entirely suitable for incorporation into the Board’s asset management system (AMS). The specific data to be compiled for turnover to the Board will be provided by the S.O. after Tender award.

G1.2.14 Model Submissions After Construction (As-Built)

- (a) The Contractor shall develop As-built models in accordance with the BIR, corresponding also to the final approved amendment plans submitted to the relevant authorities.

- (b) The Contractor shall prepare and submit two complete sets of As-Built Models and Drawings. Each complete set shall comprise the following:
 - (i) 1 soft copy of As-Built BIM Models (Native File Format) updated with S.O approved contractor's final design and embedded with COBie data to all elements identified by the Board as Facility Management (FM) and Operation & Maintenance items;
 - (ii) COBie exported Spreadsheets (Excel Spreadsheet .xls);
 - (iii) Updated database of P&ID data;
 - (iv) 1 set of A1 size paper prints (drawings for statutory submissions shall be endorsed by the appropriate Professional Engineer, Singapore); and
 - (v) 1 soft copy of the As-Built Models and Drawings in ".dgn" format.
 - (vi) 1 soft copy of the As-Built Model in ".i.dgn" format
 - (vii) 1 soft copy of the As-Built Model in ".imodel" format
 - (viii) 1 soft copy of the As-Built Model in Navisworks ".NWD" format
- (c) As-Built BIM Models and Drawings shall be endorsed "As-Built" and one set of original plotted drawings and submitted together with their soft copy in DVD-R media. All As-Built Models and Drawings in both hard and soft copies shall be submitted prior to Substantial Completion of the Works.
- (d) As-Built BIM Models and Drawings will undergo a full review by the S.O. to ensure the As-Built data meets all requirements. Upon acceptance/approval, the As-Built BIM Models/Drawings and their storage media shall become the sole property of the Board. All related costs associated with complying with these requirements is deemed to have been allowed for in the tender, whether indicated explicitly or otherwise.

G1.2.15 Integration with Asset Management System (AMS)

- (a) The Board has set up an AMS for the whole organisation and, in addition to the BIM data, data shall be provided by the Contractor for AMS Integration.
- (b) The Contractor shall provide a comprehensive computerised asset listing data/information for data integration to the Board's AMS before end of the Contract.
- (c) The Contractor will be provided with a soft copy of the latest templates used by the Board, which will include, but not limited to, the asset master list (e.g. description, asset ID, runtime, type, location, model, serial number, brand, design capacity, cost, date, warranty, spares list, checklist, and maintenance schedules). The existing templates are provided in the tender document for reference. The Contractor shall include in his rates and prices to populate and submit all the required information in the list until successful integration to Board's AMS and to the satisfaction of the S.O..

G1.2.16 Facility Management (FM) and Operations & Maintenance (O&M)

- (a) All operation and maintenance information for the Works shall be input into a database in editable formats by the Contractor. This information will be linked to and be accessed using a variety of portals. For example: the as-built BIM model, SCADA, PC/tablet based browsers, and apps.
- (b) Operation and maintenance information shall be provided in a range of formats accessible from the database(s). These shall include, but not be limited to, data fields, editable text, video and audio.
- (c) Operation and maintenance information shall include vendor information along with the appropriate documentation developed during the detailed design and construction phase of the project.
- (d) The database(s) shall have an "open architecture" that permits the expansion and inclusion of additional document types and processes that may be desired in the future without significant expenditure and allow navigation from a variety of portals.
- (e) The Facility Management (FM) information shall cover the following topics as a minimum
 - (i) Rooms, furniture, doors, windows, accessories, building services, etc. elements to report the Cobie data;
 - (ii) Maintenance report;
 - (iii) Objects Manuals;
 - (iv) Health and safety procedures, including safety regulations;
 - (v) Perform tracking on scheduled activities and warranty requirement.
- (f) The operations and maintenance information shall cover the following topics as a minimum:
 - (i) Plant operation;
 - (ii) Process control narratives;
 - (iii) Standard operating procedures;
 - (iv) Equipment maintenance;
 - (v) Equipment manuals and shop drawings;
 - (vi) Equipment and engineering data;
 - (vii) Construction information;
 - (viii) Training materials;
 - (ix) Health and safety procedures, including safety regulations, chemical material safety data sheets (MSDS), hazardous area and confined space details, confined space entry procedures and lockout/tagout procedures;
 - (x) Reference information (process theory, chemical interactions); and
 - (xi) Maintenance information.

G1.2.17 Information Management Roles

- (a) The Contractor shall have the following staff as a minimum to successfully manage the information:
 - (i) Information Manager - Responsible for the overall setup and configuration of the Contractor CDE, the BIM Execution Plan and a single point of contact for the S.O.. The Information Manager must have about 10 years' experience in the management of information management processes, including setup and management of the CDE and interaction with the BIM Authoring Tools given in Table G1.2-2.
 - (ii) Database Lead: Responsible for setup and configuration of the P&ID database and provides support during the Contractor design and construction of MEICA elements. Must be proficient in Bentley Open Plant P&ID, SQL and have about 5 years' experience in this field.
 - (iii) Building Information Management Manager (BIMM) – the BIMM shall manage and coordinate the overall use of BIM for the project. The BIMM shall have about 5 years of relevant experience in the implementation and management of CAD and 3D models for similar scale and complexity projects, of which about five (5) years of experience shall be in development and implementation of Building Information Management (BIM) systems. In addition, he shall have a comprehensive knowledge of related BIM technologies to lead and/or assist in the creation of BIM deliverables such as BIM Execution Plans, Clash Detection Reports, BIM Quality Control Plans, BIM Design Review Procedures, Construction Sequencing (4D) and Cost Estimating (5D). The BIM Manager shall have relevant engineering degree or certification. The BIMM shall be expert in the BIM Authoring Tools given in Table G1.2-2.
 - (iv) BIM Authoring Leads: Responsible for the ongoing updating of the model to construction and as-built status. Must be proficient and have at about 5 years' experience of working with the BIM Authoring Tools given in Table G1.2-2.
 - (v) Document Controllers: Responsible for the transmission of the content from the Contractor's CDE to the TWRP CDE via the project-wide EDMS.

END OF SECTION